

116. Introduction to 200 Gb/s and 400 Gb/s networks

116.1 Overview

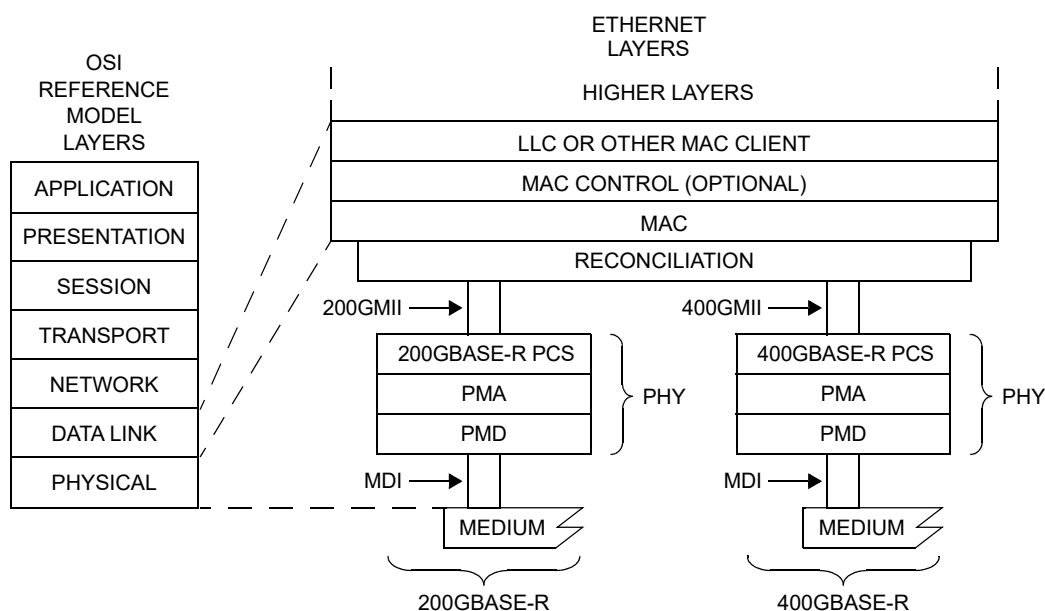
116.1.1 Scope

This clause describes the general requirements for 200 Gigabit and 400 Gigabit Ethernet.

200 Gigabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 200 Gb/s, coupled with any IEEE 802.3 200GBASE Physical Layer implementation and is defined for full duplex operation only. 400 Gigabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 400 Gb/s, coupled with any IEEE 802.3 400GBASE Physical Layer implementation and is defined for full duplex operation only.

116.1.2 Relationship of 200 Gigabit and 400 Gigabit Ethernet to the ISO OSI reference model

200 Gigabit and 400 Gigabit Ethernet couples the IEEE 802.3 MAC to a family of 200 Gb/s and 400 Gb/s Physical Layer devices. The relationships among 200 Gigabit and 400 Gigabit Ethernet, the IEEE 802.3 MAC, and the ISO Open System Interconnection (OSI) reference model are shown in Figure 116–1.



200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 116–1—Architectural positioning of 200 Gigabit and 400 Gigabit Ethernet

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience. The only exceptions are as follows:

- a) The 200GMII and 400GMII, which, when implemented as logical interconnection points between the MAC sublayer and the Physical Layer device (PHY), use a 64-bit wide data path as specified in Clause 117. Physical instantiations of this interface may use other data-path widths.
- b) The management interface, which, when physically implemented as the MDIO/MDC (Management Data Input/Output and Management Data Clock) at an observable interconnection port, uses a bit-wide data path as specified in Clause 45.
- c) The PMA service interface, which, when physically implemented as 400GAUI-16 (400 Gb/s sixteen-lane Attachment Unit Interface) at an observable interconnection port, uses a 16-lane data path as specified in Annex 120B or Annex 120C.
- d) The PMA service interface, which, when physically implemented as 200GAUI-8 (200 Gb/s eight-lane Attachment Unit Interface) or 400GAUI-8 (400 Gb/s eight-lane Attachment Unit Interface) at an observable interconnection port, uses an 8-lane data path as specified in Annex 120B, Annex 120C, Annex 120D, or Annex 120E.
- e) The PMA service interface, which, when physically implemented as 200GAUI-4 (200 Gb/s four-lane Attachment Unit Interface) at an observable interconnection port, uses a 4-lane data path as specified in Annex 120D or Annex 120E.
- f) The MDI as specified in Clause 123 for 400GBASE-SR16 uses a 16-lane data path.
- g) The MDIs as specified in:
 - Clause 122 for 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8
 - Clause 138 for 400GBASE-SR8
 - Clause 150 for 400GBASE-SR4.2all use an 8-lane data path.
- h) The MDIs as specified in:
 - Clause 121 for 200GBASE-DR4
 - Clause 122 for 200GBASE-FR4, 200GBASE-LR4, and 200GBASE-ER4
 - Clause 124 for 400GBASE-DR4
 - Clause 136 for 200GBASE-CR4
 - Clause 137 for 200GBASE-KR4
 - Clause 138 for 200GBASE-SR4
 - Clause 151 for 400GBASE-FR4 and 400GBASE-LR4-6all use a 4-lane data path.

116.1.3 Nomenclature

The nomenclature employed by the 200 Gb/s and 400 Gb/s Physical Layer is explained as follows.

The alpha-numeric prefix 200GBASE in the port type (e.g., 200GBASE-R) represents a family of Physical Layer devices operating at a speed of 200 Gb/s. The alpha-numeric prefix 400GBASE in the port type (e.g., 400GBASE-R) represents a family of Physical Layer devices operating at a speed of 400 Gb/s.

200GBASE-R represents a family of Physical Layer devices using the Physical Coding Sublayer for 200 Gb/s operation over multiple PCS lanes (see Clause 119). Physical Layer devices listed in Table 116–1 are defined for operation at 200 Gb/s.

400GBASE-R represents a family of Physical Layer devices using the Physical Coding Sublayer for 400 Gb/s operation over multiple PCS lanes (see Clause 119). Physical Layer devices listed in Table 116–2 are defined for operation at 400 Gb/s.

Table 116–1—200 Gb/s PHYs

Name	Description
200GBASE-KR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of an electrical backplane (see Clause 137)
200GBASE-CR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of twinaxial copper cable (see 1.4.559 and Clause 136)
200GBASE-SR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of multimode fiber (see Clause 138)
200GBASE-DR4	200 Gb/s PHY using 200GBASE-R encoding over four lanes of single-mode fiber, with reach up to at least 500 m (see Clause 121)
200GBASE-FR4	200 Gb/s PHY using 200GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 2 km (see Clause 122)
200GBASE-LR4	200 Gb/s PHY using 200GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 10 km (see Clause 122)
200GBASE-ER4	200 Gb/s PHY using 200GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 40 km (see Clause 122)

Table 116–2—400 Gb/s PHYs

Name	Description
400GBASE-SR16	400 Gb/s PHY using 400GBASE-R encoding over sixteen lanes of multimode fiber, with reach up to at least 100 m (see Clause 123)
400GBASE-SR8	400 Gb/s PHY using 400GBASE-R encoding over eight lanes of multimode fiber, with reach up to at least 100 m (see Clause 138)
400GBASE-SR4.2	400 Gb/s PHY using 400GBASE-R encoding over eight lanes on multimode fiber in a bidirectional WDM format, with reach up to at least 150 m (see Clause 150)
400GBASE-DR4	400 Gb/s PHY using 400GBASE-R encoding over four lanes of single-mode fiber, with reach up to at least 500 m (see Clause 124)
400GBASE-FR8	400 Gb/s PHY using 400GBASE-R encoding over eight WDM lanes on single-mode fiber, with reach up to at least 2 km (see Clause 122)
400GBASE-FR4	400 Gb/s PHY using 400GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 2 km (see Clause 151)
400GBASE-LR4-6	400 Gb/s PHY using 400GBASE-R encoding over four WDM lanes on single-mode fiber, with reach up to at least 6 km (see Clause 151)
400GBASE-LR8	400 Gb/s PHY using 400GBASE-R encoding over eight WDM lanes on single-mode fiber, with reach up to at least 10 km (see Clause 122)
400GBASE-ER8	400 Gb/s PHY using 400GBASE-R encoding over eight WDM lanes on single-mode fiber, with reach up to at least 40 km (see Clause 122)

116.1.4 Physical Layer signaling systems

This standard specifies a family of Physical Layer implementations. Table 116–3, Table 116–4, and Table 116–5 specify the correlation between PHY types and clauses. Implementations conforming to one or more PHY types meet the requirements of the corresponding clauses.

Table 116–3—PHY type and clause correlation (200GBASE copper)

PHY type	Clause ^a										
	73	78	117		118	119	120	120B	120D	136	137
	Auto-Negotiation	EEE	RS	200GMII	200GMII Extender	200GBASE-R PCS	200GBASE-R PMA	200GAUI-8 C2C	200GAUI-4 C2C	200GBASE-CR4 PMD	200GBASE-KR4 PMD
200GBASE-KR4	M	O	M	O	O	M	M	O	O		M
200GBASE-CR4	M	O	M	O	O	M	M	O	O	M	

^aO = Optional, M = Mandatory.

Table 116–4—PHY type and clause correlation (200GBASE optical)

PHY type	Clause ^a														
	78	117		118	119	120	120B	120C	120D	120E	121	122			138
	EEE	RS	200GMII	200GMII Extender	200GBASE-R PCS	200GBASE-R PMA	200GAUI-8 C2C	200GAUI-8 C2M	200GAUI-4 C2C	200GAUI-4 C2M	200GBASE-DR4 PMD	200GBASE-FR4 PMD	200GBASE-LR4 PMD	200GBASE-ER4 PMD	200GBASE-SR4 PMD
200GBASE-SR4	O	M	O	O	M	M	O	O	O	O					M
200GBASE-DR4	O	M	O	O	M	M	O	O	O	O	M				
200GBASE-FR4	O	M	O	O	M	M	O	O	O	O		M			
200GBASE-LR4	O	M	O	O	M	M	O	O	O	O			M		
200GBASE-ER4	O	M	O	O	M	M	O	O	O	O				M	

^a O = Optional, M = Mandatory.

Table 116–5—PHY type and clause correlation (400GBASE optical)

PHY type	Clause ^a																		
	78	117		118	119	120	120B	120C	120D	120E	123	138	150	124	122	151		122	
	EEE	RS	400GMII	400GMII Extender	400GBASE-R PCS	400GBASE-R PMA	400GAUI-16 C2C	400GAUI-16 C2M	400GAUI-8 C2C	400GAUI-8 C2M	400GBASE-SR16 PMD	400GBASE-SR8 PMD	400GBASE-SR4.2 PMD	400GBASE-DR4 PMD	400GBASE-FR8 PMD	400GBASE-FR4 PMD	400GBASE-LR4-6 PMD	400GBASE-LR8 PMD	400GBASE-ER8 PMD
400GBASE-SR16	O	M	O	O	M	M	O	O	O	O	M								
400GBASE-SR8	O	M	O	O	M	M	O	O	O	O		M							
400GBASE-SR4.2	O	M	O	O	M	M	O	O	O	O			M						
400GBASE-DR4	O	M	O	O	M	M	O	O	O	O				M					
400GBASE-FR8	O	M	O	O	M	M	O	O	O	O					M				
400GBASE-FR4	O	M	O	O	M	M	O	O	O	O						M			
400GBASE-LR4-6	O	M	O	O	M	M	O	O	O	O							M		
400GBASE-LR8	O	M	O	O	M	M	O	O	O	O								M	
400GBASE-ER8	O	M	O	O	M	M	O	O	O	O									M

^a O = Optional, M = Mandatory.

116.2 Summary of 200 Gigabit and 400 Gigabit Ethernet sublayers

116.2.1 Reconciliation Sublayer (RS) and Media Independent Interface

The Media Independent Interface (Clause 117) provides a logical interconnection between the MAC sublayer and Physical Layer devices (PHYs). The Media Independent Interface is not intended to be physically instantiated, rather it can logically connect layers within a device.

The 200GMII supports 200 Gb/s operation and the 400GMII supports 400 Gb/s operation through its 64-bit-wide transmit and receive data paths. The Reconciliation Sublayer (RS) provides a mapping between the signals provided at the Media Independent Interface (200GMII and 400GMII) and the MAC/PLS service definition.

While the 200GMII and 400GMII are optional interfaces, they are used extensively in this standard as a basis for functional specification and provides a common service interface for the physical coding sublayer defined in Clause 119.

116.2.2 200GMII and 400GMII Extender Sublayers (200GXS and 400GXS)

The 200 Gigabit 200GMII Extender Sublayer (200GXS) is part of the 200GMII Extender (Clause 118). It is identical in function to the 200GBASE-R PCS in Clause 119 with the exceptions defined in Clause 118. The 400 Gigabit 400GMII Extender Sublayer (400GXS) is part of the 400GMII Extender (Clause 118). It is identical in function to the 400GBASE-R PCS in Clause 119 with the exceptions defined in Clause 118.

The optional 200GMII Extender (Clause 118) can be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 200GMII. The optional 400GMII Extender (Clause 118) can be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 400GMII.

116.2.3 Physical Coding Sublayer (PCS)

The terms 200GBASE-R and 400GBASE-R refer to a specific family of Physical Layer implementations based upon the 64B/66B data coding method specified in Clause 119 and the PMA specifications defined in Clause 120. Clause 119 PCSs perform encoding (decoding) of data from (to) the 200GMII or 400GMII to 256B/257B code blocks, apply FEC, distribute the data to multiple lanes, and transfer the encoded data to the PMA.

The 200GBASE-R PCS has almost the same functionality as the 200GXS, and therefore may be configured as a 200GXS in order to implement part of the optional 200GMII Extender (see Clause 118). The 400GBASE-R PCS has almost the same functionality as the 400GXS, and therefore may be configured as a 400GXS in order to implement part of the optional 400GMII Extender (see Clause 118).

116.2.4 Physical Medium Attachment (PMA) sublayer

The PMA provides a medium-independent means for the PCS to support the use of a range of physical media. The 200GBASE-R and 400GBASE-R PMAs perform the mapping of transmit and receive data streams between the PCS and PMA via the PMA service interface, and the mapping and multiplexing of transmit and receive data streams between the PMA and PMD via the PMD service interface. In addition, the PMA performs retiming of the received data stream when appropriate, optionally provides data loopback at the PMA or PMD service interface, and optionally provides test pattern generation and checking.

The 200GBASE-R and 400GBASE-R PMAs are specified in Clause 120.

116.2.5 Physical Medium Dependent (PMD) sublayer

The Physical Medium Dependent sublayer is responsible for interfacing to the transmission medium. The PMD is located just above the Medium Dependent Interface (MDI). The MDI, logically subsumed within each PMD subclause, is the actual medium attachment for the various supported media.

The 200GBASE-R PMDs and their corresponding media are specified in Clause 121, Clause 122, and Clause 136 through Clause 138. The 400GBASE-R PMDs and their corresponding media are specified in Clause 122 through Clause 124, Clause 138, and Clause 150.

116.2.6 Management interface (MDIO/MDC)

The optional MDIO/MDC management interface (Clause 45) provides an interconnection between MDIO Manageable Devices (MMDs) and Station Management (STA) entities.

116.2.7 Management

Managed objects, attributes, and actions are defined for all 200 Gigabit and 400 Gigabit Ethernet components. These items are defined in Clause 30.

116.3 Service interface specification method and notation

The service interface specification for the 200GBASE-R and 400GBASE-R Physical Layers is as per the definition in 1.2.2. Note that the 200GBASE-R and 400GBASE-R inter-sublayer service interfaces use

multiple scalar REQUEST and INDICATION primitives, to indicate the transfer of multiple independent streams of data units, as defined in 116.3.1 through 116.3.3.

116.3.1 Inter-sublayer service interface

The inter-sublayer service interface is described in an abstract manner and does not imply any particular implementation. The inter-sublayer service interface primitives are defined as follows:

IS_UNITDATA_*i*.request
IS_UNITDATA_*i*.indication
IS_SIGNAL.indication

The IS_UNITDATA_*i*.request (where $i = 0$ to $n - 1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next lower (closer to the medium) sublayer. The IS_UNITDATA_*i*.indication (where $i = 0$ to $n - 1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next higher (closer to the MAC) sublayer. The IS_SIGNAL.indication primitive is used to define the transfer of signal status from a sublayer to the next higher sublayer.

116.3.2 Instances of the Inter-sublayer service interface

The inter-sublayer interface can be instantiated between different sublayers, hence a prefix notation is defined to identify a specific instance of an inter-sublayer service interface. The following prefixes are defined:

- a) PMD:—for primitives issued on the interface between the PMD sublayer and the PMA sublayer called the PMD service interface.
- b) PMA:—for primitives issued on the interface between the PMA sublayer and the PCS, DTE 200GXS, DTE 400GXS, or PMA sublayer above called the PMA service interface.
- c) PHY XS:—for primitives issued on the interface between the PMA sublayer and the PHY 200GXS sublayer or PHY 400GXS sublayer called the PHY XS service interface.

Examples of inter-sublayer service interfaces for 200GBASE-R and 400GBASE-R with their corresponding instance names are illustrated in Figure 116–2 and Figure 116–3, respectively. For example, the primitives for one instance of the inter-sublayer service interface, named the PMD service interface, are identified as follows:

PMD:IS_UNITDATA_*i*.request
PMD:IS_UNITDATA_*i*.indication
PMD:IS_SIGNAL.indication.

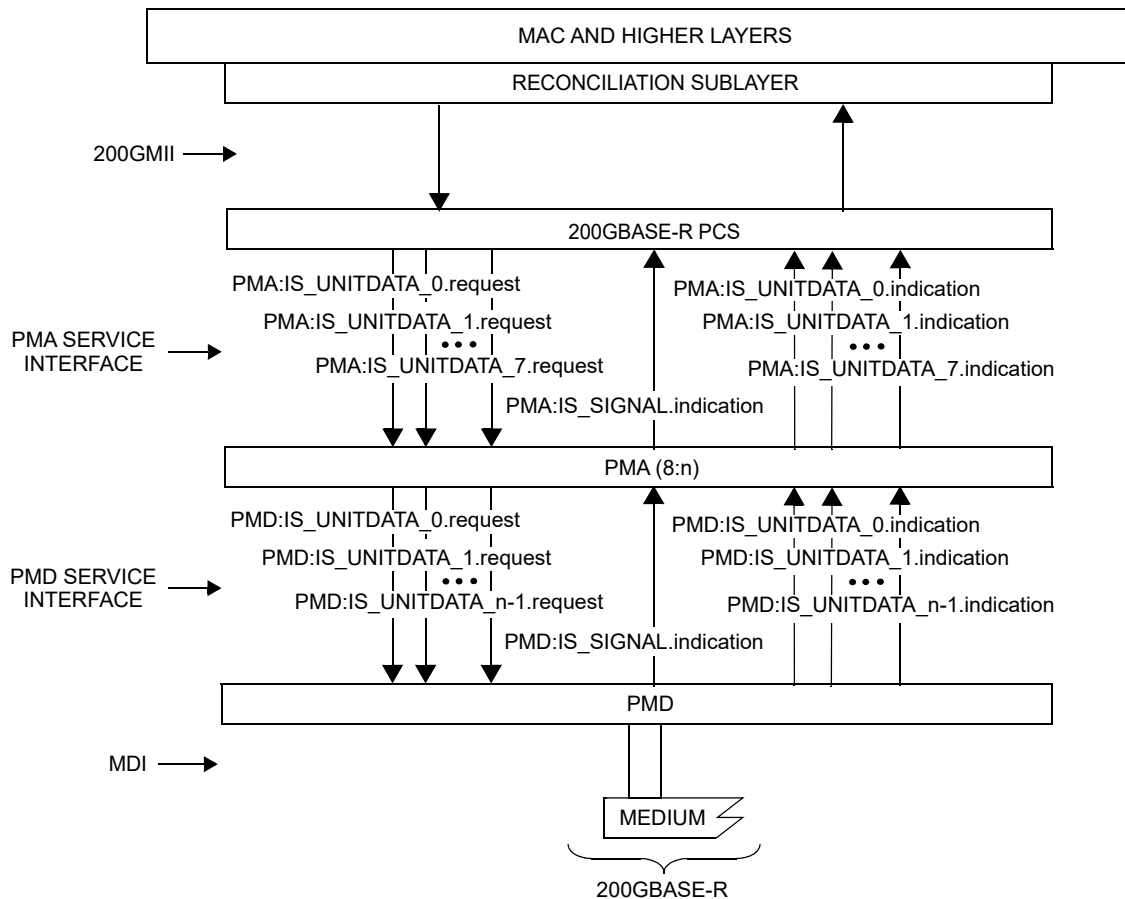
Primitives for other instances, of inter-sublayer interfaces, are represented in a similar manner as described above.

116.3.3 Semantics of inter-sublayer service interface primitives

The semantics of the inter-sublayer service interface primitives for the 200GBASE-R and 400GBASE-R sublayers are described in 116.3.3.1 through 116.3.3.3.

116.3.3.1 IS_UNITDATA_*i*.request

The IS_UNITDATA_*i*.request (where $i = 0$ to $n - 1$) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next lower sublayer, where n is the number of parallel streams of data units.



200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

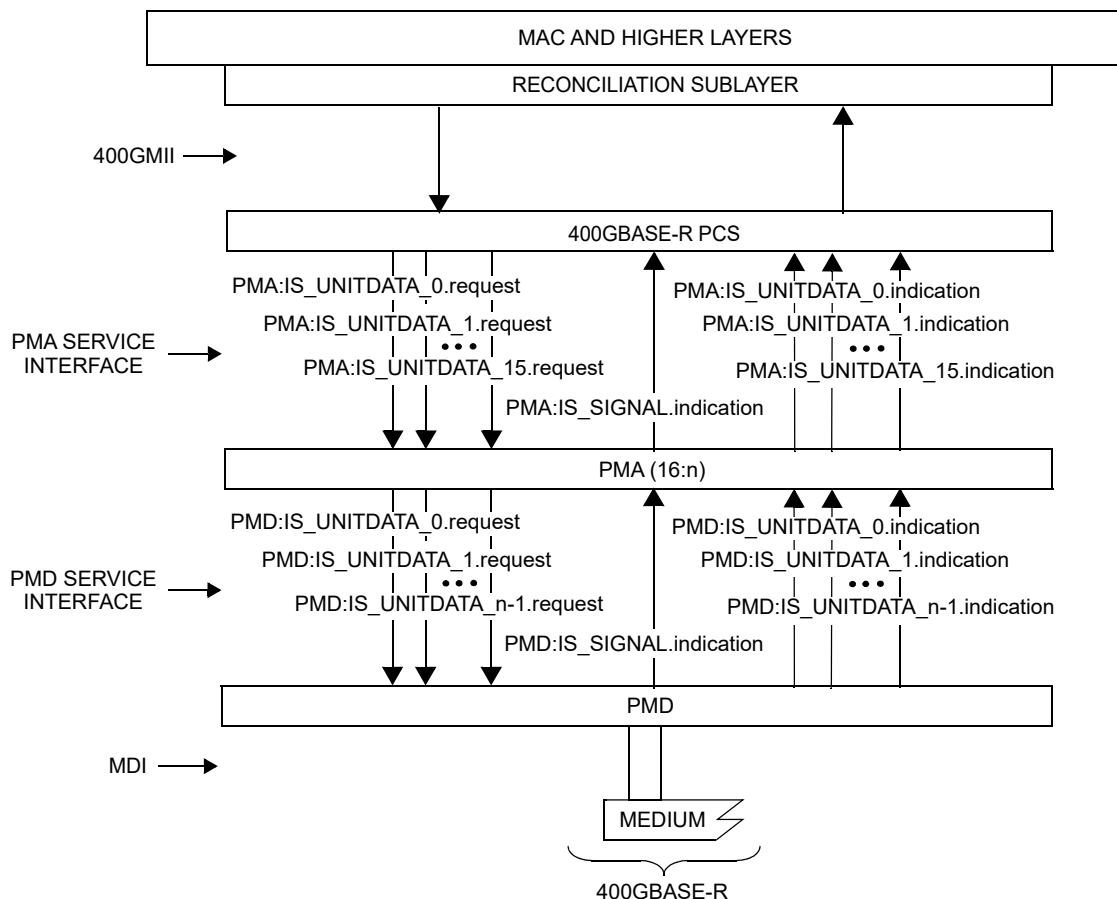
n = NUMBER OF PARALLEL STREAMS OF DATA UNITS

Figure 116-2—200GBASE-R inter-sublayer service interfaces

116.3.3.1.1 Semantics of the service primitive

```
IS_UNITDATA_0.request(tx_symbol)
IS_UNITDATA_1.request(tx_symbol)
...
IS_UNITDATA_n-1.request(tx_symbol)
```

The data conveyed by IS_UNITDATA_0.request to IS_UNITDATA_n-1.request consists of n parallel continuous streams of encoded symbols, one stream for each lane. Depending on the specific instance of the inter-sublayer service interface each of the tx_symbol parameters can either take one of two values: zero or one; or take one of four values: zero, one, two, or three.



400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
n = NUMBER OF PARALLEL STREAMS OF DATA UNITS

Figure 116-3—400GBASE-R inter-sublayer service interfaces

116.3.3.1.2 When generated

The sublayer continuously sends n parallel symbol streams $IS_UNITDATA_i.request(tx_symbol)$ to the next lower sublayer, each at a nominal signaling rate defined by a specific instance of the inter-sublayer service interface.

116.3.3.1.3 Effect of receipt

The effect of receipt of this primitive is defined by the sublayer that receives this primitive.

116.3.3.2 $IS_UNITDATA_i.indication$

The $IS_UNITDATA_i.indication$ (where $i = 0$ to $n - 1$) primitive is used to define the transfer of multiple streams of data units from the sublayer to the next higher sublayer, where n is the number of parallel streams of data units.

116.3.3.2.1 Semantics of the service primitive

IS_UNITDATA_0.indication(rx_symbol)
IS_UNITDATA_1.indication(rx_symbol)
...
IS_UNITDATA_n-1.indication(rx_symbol)

The data conveyed by IS_UNITDATA_0.indication to IS_UNITDATA_n-1.indication consists of n parallel continuous streams of encoded symbols, one stream for each lane. Depending on the specific instance of the inter-sublayer service interface each of the rx_symbol parameters can either take one of two values: zero or one; or take one of four values: zero, one, two, or three.

116.3.3.2.2 When generated

The sublayer continuously sends n parallel symbol streams IS_UNITDATA_i.indication(rx_symbol) to the next higher sublayer, each at a nominal signaling rate defined by a specific instance of the inter-sublayer service interface.

116.3.3.2.3 Effect of receipt

The effect of receipt of this primitive is defined by the sublayer that receives this primitive.

116.3.3.3 IS_SIGNAL.indication

The IS_SIGNAL.indication primitive is generated by the sublayer to the next higher sublayer to indicate the status of the receive process. This primitive is generated by the receive process to propagate the detection of severe error conditions (e.g., no valid signal being received by the sublayer that generates this primitive) to the next higher sublayer.

116.3.3.3.1 Semantics of the service primitive

IS_SIGNAL.indication(SIGNAL_OK)

The SIGNAL_OK parameter can take on one of two values: OK or FAIL. A value of FAIL denotes that invalid data is being presented (rx_symbol parameters undefined) by the sublayer to the next higher sublayer. A value of OK does not guarantee valid data is being presented by the sublayer to the next higher sublayer.

116.3.3.3.2 When generated

The sublayer generates the IS_SIGNAL.indication primitive to the next higher sublayer whenever there is change in the value of the SIGNAL_OK parameter.

116.3.3.3.3 Effect of receipt

The effect of receipt of this primitive is defined by the sublayer that receives this primitive.

116.4 Delay constraints

Predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementations conform to certain delay maxima, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices.

Table 116–6 and Table 116–7 contain the values of maximum sublayer delay (sum of transmit and receive delays at one end of the link) in bit times as specified in 1.4 and pause_quanta as specified in 31B.2 for 200 Gigabit and 400 Gigabit Ethernet, respectively. If a PHY contains an Auto-Negotiation sublayer, the delay of the Auto-Negotiation sublayer is included within the delay of the PMD and medium.

Table 116–6—Sublayer delay constraints (200GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quanta) ^b	Maximum (ns)	Notes ^c
200G MAC, RS, and MAC Control	49 152	96	245.76	See 117.1.4.
200GBASE-R PCS or 200GXS ^d	160 256	313	801.28	See 119.5.
200GBASE-R PMA	18 432	36	92.16	See 120.5.4.
200GBASE-KR4 PMD	8 192	16	40.96	Includes allocation of 20 ns for one direction through backplane medium. See 137.5.
200GBASE-CR4 PMD	8 192	16	40.96	Includes allocation of 20 ns for one direction through cable medium. See 137.5.
200GBASE-SR4 PMD	4 096	8	20.48	See 138.3.
200GBASE-DR4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 121.3.1.
200GBASE-FR4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 122.3.1.
200GBASE-LR4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 122.3.1.
200GBASE-ER4 PMD	4 096	8	20.48	Includes 2 m of fiber. See 122.3.1.

^a For 200GBASE-R, 1 bit time (BT) is equal to 5 ps. (See 1.4.215 for the definition of bit time.)

^b For 200GBASE-R, 1 pause_quantum is equal to 2.56 ns. (See 31B.2 for the definition of pause_quanta.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^d If an implementation includes the 200GMII extender, the delay associated with the 200GMII extender includes two 200GXS sublayers.

Table 116–7—Sublayer delay constraints (400GBASE)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quanta) ^b	Maximum (ns)	Notes ^c
400G MAC, RS, and MAC Control	98 304	192	245.76	See 117.1.4.
400GBASE-R PCS or 400GXS ^d	320 000	625	800	See 119.5.
400GBASE-R PMA	36 864	72	92.16	See 120.5.4.
400GBASE-SR16 PMD	8 192	16	20.48	Includes 2 m of fiber. See 123.3.1.
400GBASE-SR8 PMD	8 192	16	20.48	Includes 2 m of fiber. See 138.3.1.
400GBASE-SR4.2 PMD	8 192	16	20.48	Includes 2 m of fiber. See 150.3.1.
400GBASE-DR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 124.3.1.

Table 116–7—Sublayer delay constraints (400GBASE) (*continued*)

Sublayer	Maximum (bit time) ^a	Maximum (pause_quantum) ^b	Maximum (ns)	Notes ^c
400GBASE-FR8 PMD	8 192	16	20.48	Includes 2 m of fiber. See 122.3.1.
400GBASE-FR4 PMD	8 192	16	20.48	Includes 2 m of fiber. See 151.3.1.
400GBASE-LR4-6 PMD	8 192	16	20.48	Includes 2 m of fiber. See 151.3.1.
400GBASE-LR8 PMD	8 192	16	20.48	Includes 2 m of fiber. See 122.3.1.
400GBASE-ER8 PMD	8 192	16	20.48	Includes 2 m of fiber. See 122.3.1.

^a For 400GBASE-R, 1 bit time (BT) is equal to 2.5 ps. (See 1.4.215 for the definition of bit time.)

^b For 400GBASE-R, 1 pause_quantum is equal to 1.28 ns. (See 31B.2 for the definition of pause_quantum.)

^c Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^d If an implementation includes the 400GMII extender, the delay associated with the 400GMII extender includes two 400GXS sublayers.

See 80.4 for the calculation of bit time per meter of fiber or electrical cable.

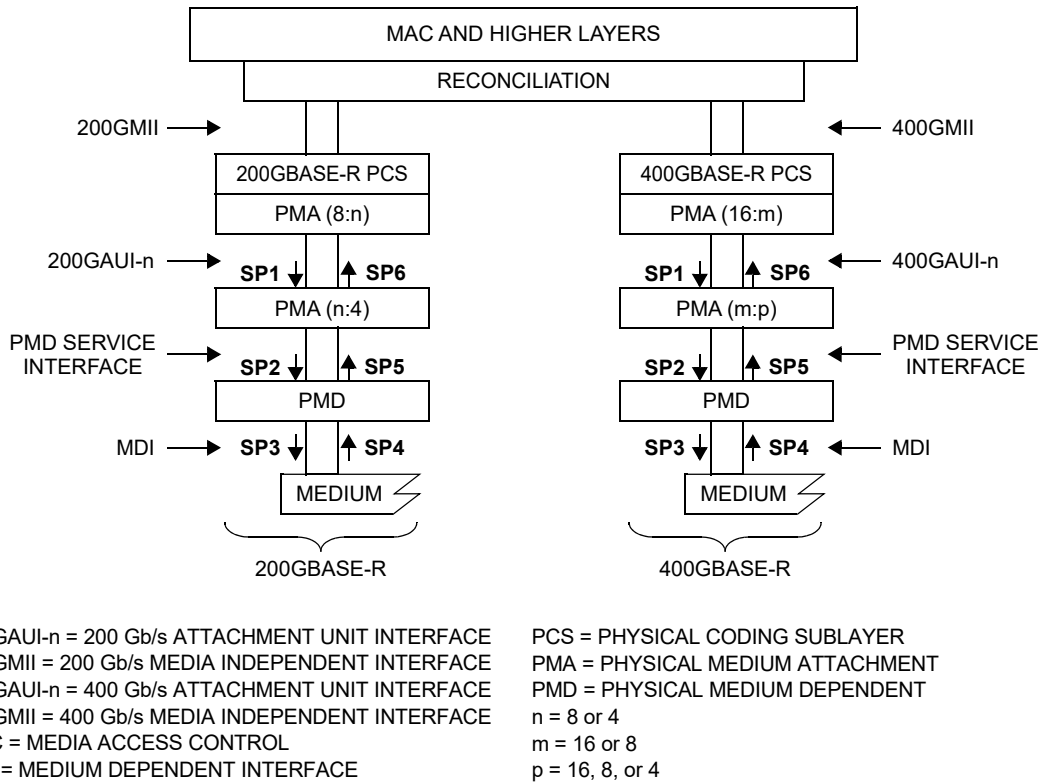
See 31B.3.7 for PAUSE reaction timing constraints for stations at operating speeds of 200 Gb/s and 400 Gb/s.

116.5 Skew constraints

Skew (or relative delay) can be introduced between lanes by both active and passive elements of a 200GBASE-R or 400GBASE-R link. Skew is defined as the difference between the times of the earliest PCS lane and latest PCS lane for the one to zero transition of the alignment marker sync bits. The PCS deskew function (see 119.2.5.1) compensates for all lane-to-lane Skew observed at the receiver. The Skew between the lanes is kept within limits as shown in Table 116–8 so that the transmitted information on the lanes can be reassembled by the receive PCS.

Skew Variation may be introduced due to variations in electrical, thermal or environmental characteristics. Skew Variation is defined as the change in Skew between any PCS lane and any other PCS lane over the entire time that the link is in operation. From the time the link is brought up, Skew Variation is limited so that each PCS lane always traverses the same lane between any pair of adjacent sublayers while the link remains in operation.

The maximum Skew and Skew Variation at physically instantiated interfaces is specified at Skew points SP1, SP2, and SP3 for the transmit direction and SP4, SP5, and SP6 for the receive direction as illustrated in Figure 116–4 (single 200GAUI-n or 400GAUI-n interface) and Figure 116–5 (multiple 200GAUI-n or 400GAUI-n interfaces).



**Figure 116-4—200GBASE-R and 400GBASE-R Skew points
for single 200GAUI-n or 400GAUI-n**

In the transmit direction, the Skew points are defined in the following locations:

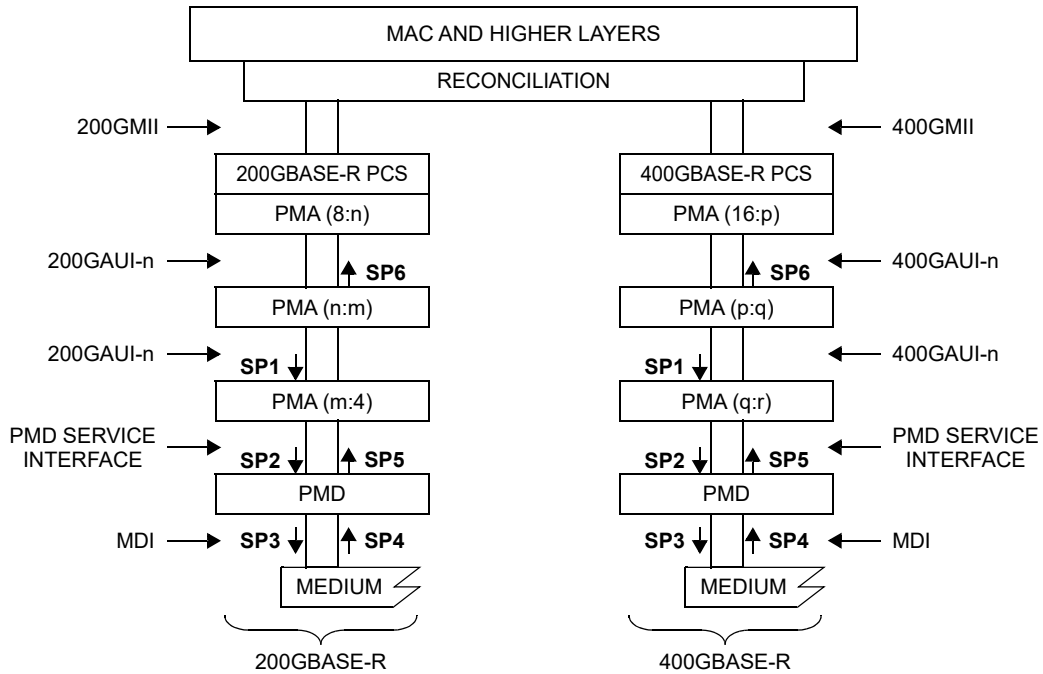
- SP1 on the 200GAUI-n/400GAUI-n interface, at the input of the PMA closest to the PMD;
- SP2 on the PMD service interface, at the input of the PMD;
- SP3 at the output of the PMD, at the MDI.

In the receive direction, the Skew points are defined in the following locations:

- SP4 at the MDI, at the input of the PMD;
- SP5 on the PMD service interface, at the output of the PMD;
- SP6 on the 200GAUI-n/400GAUI-n interface, at the output of the PMA closest to the 200GBASE-R/400GBASE-R PCS or DTE 200GXS/400GXS.

The allowable limits for Skew are shown in Table 116-8 and the allowable limits for Skew Variation are shown in Table 116-9.

The Skew requirements for the PCS, PMA and PMD sublayers are specified in the respective clauses as noted in Table 116-8 and Table 116-9.



200GAUI-n = 200 Gb/s ATTACHMENT UNIT INTERFACE
200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
400GAUI-n = 400 Gb/s ATTACHMENT UNIT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
n = 8 or 4
m = 8 or 4
p = 16 or 8
q = 16 or 8
r = 16, 8, or 4

**Figure 116-5—200GBASE-R and 400GBASE-R Skew points
for multiple 200GAUI-n or 400GAUI-n**

Table 116–8—Summary of Skew constraints

Skew points	Maximum Skew (ns) ^a	Maximum Skew for 200GBASE-R or 400GBASE-R PCS lane (UI) ^b	Notes ^c
SP1	29	≈ 770	See 120.5.3.1
SP2	43	≈ 1142	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP3	54	≈ 1434	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP4	134	≈ 3559	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP5	145	≈ 3852	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP6	160	≈ 4250	See 120.5.3.5
At PCS receive	180	≈ 4781	See 119.2.5.1

^a The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^b The symbol ≈ indicates approximate equivalent of maximum Skew in UI based on 1 UI equals 37.64706 ps at PCS lane signaling rate of 26.5625 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

Table 116–9—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation (ns)	Maximum Skew Variation for 26.5625 GBd PMD lane (UI) ^a	Maximum Skew Variation for 53.125 GBd PMD lane (UI) ^b	Notes ^c
SP1	0.2	≈ 5	N/A	See 120.5.3.1
SP2	0.4	≈ 11	≈ 21	See 120.5.3.3, 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP3	0.6	≈ 16	≈ 32	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP4	3.4	≈ 90	≈ 181	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP5	3.6	≈ 96	≈ 191	See 121.3.2, 122.3.2, 123.3.2, 124.3.2, 136.6.2, 137.6.2, 138.3.2.2, 150.3.2, or 151.3.2
SP6	3.8	≈ 101	N/A	See 120.5.3.5
At PCS receive	4	≈ 106	N/A	See 119.2.5.1

^a The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 37.64706 ps at PMD lane signaling rate of 26.5625 GBd.

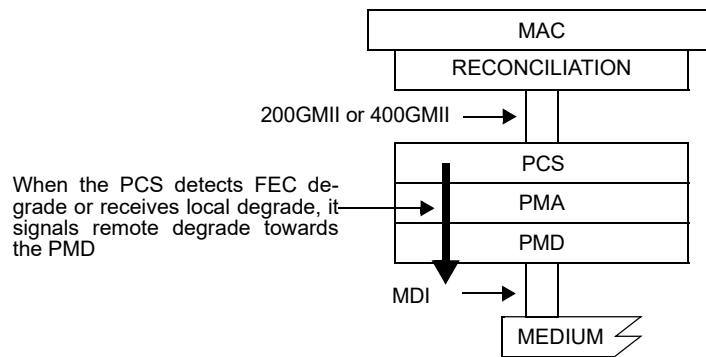
^b The symbol $\approx$ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^c Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

116.6 FEC Degrad

FEC degrade is a feature allowing for the optional detection of a non-service affecting link degradation condition based on exceeding a threshold for FEC corrected errors. If there are multiple FEC decoders in a given direction of transmission between the MAC sublayers at each end of the link, a Local Degrade condition is cascaded in that direction of transmission to convey the fact that one or more FEC decoders in the path have exceeded their threshold of FEC corrected errors. If any FEC decoder in a given direction of transmission exceeds its provisioned threshold for FEC corrected errors, a Remote Degrade condition is indicated in the opposite direction of transmission from the PCS or XS closest to the MAC.

Figure 116–6 illustrates the signaling of the Remote Degraded condition in the case that there is no Clause 118 extender sublayer present between the MAC and the PCS. Note that the PCS will not initiate the signaling for local degrade in this configuration as there are no additional FEC decoders in the receive direction between the PCS and the MAC.



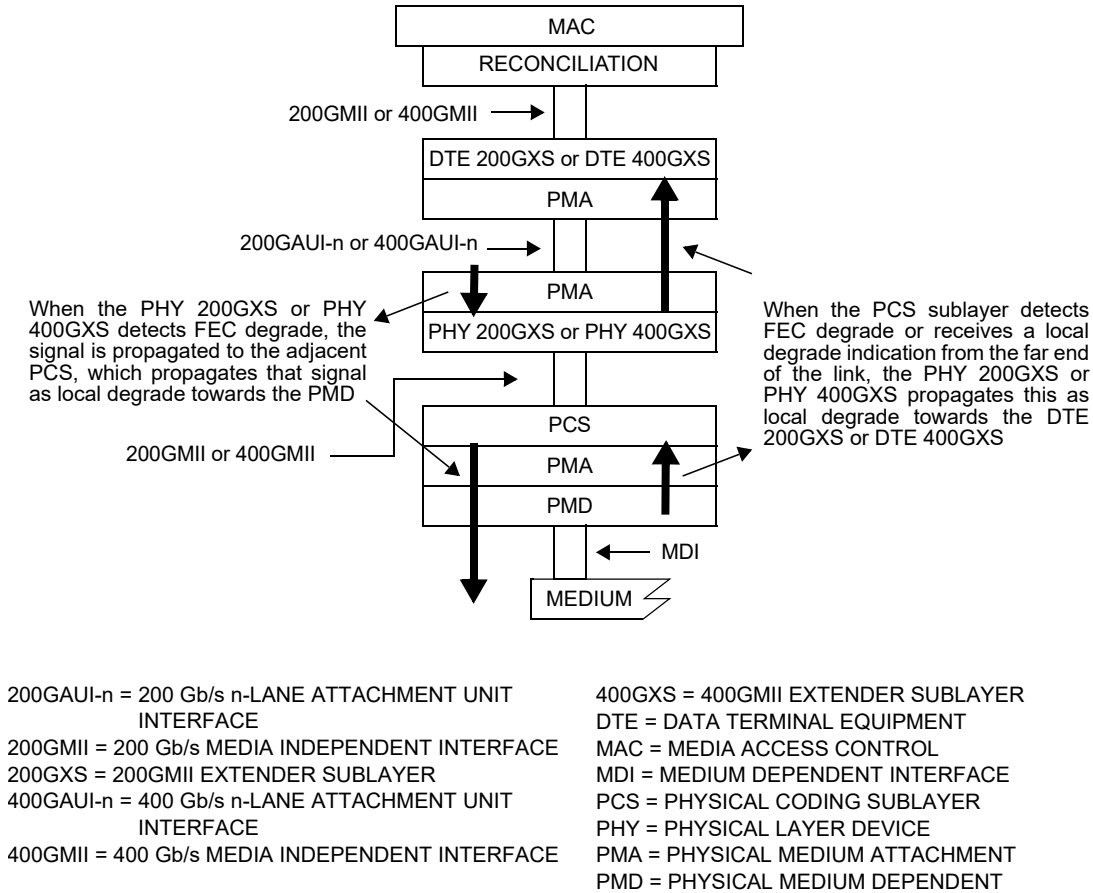
200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
DTE = DATA TERMINAL EQUIPMENT
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 116–6—Remote Degrade signaling without Extender Sublayer

Figure 116–7 illustrates the signaling of the Local Degraded condition in the case that a Clause 118 extender sublayer is present between the MAC and the PCS.

Figure 116–8 illustrates the signaling of the Remote Degraded condition in the case that a Clause 118 extender sublayer is present between the MAC and the PCS.



200GAUI-n = 200 Gb/s n-LANE ATTACHMENT UNIT
INTERFACE
200GMII = 200 Gb/s MEDIA INDEPENDENT INTERFACE
200GXS = 200GMII EXTENDER SUBLAYER
400GAUI-n = 400 Gb/s n-LANE ATTACHMENT UNIT
INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE

400GXS = 400GMII EXTENDER SUBLAYER
DTE = DATA TERMINAL EQUIPMENT
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 116-7—Local Degrad Signaling with Extender Sublayer

116.7 State diagrams

State diagrams take precedence over text.

The conventions of 1.2 are adopted, along with the extensions listed in 21.5.

Multiple states of a function that have a transition to a common state utilizing different qualifiers (for example, multiple exit conditions to an IDLE or WAIT state) may be indicated by a shared arrow. An exit transition arrow connects to the shared arrow, and the qualifier is met prior to termination of the transition arrow on the shared arrow. The shared arrow has no qualifier.

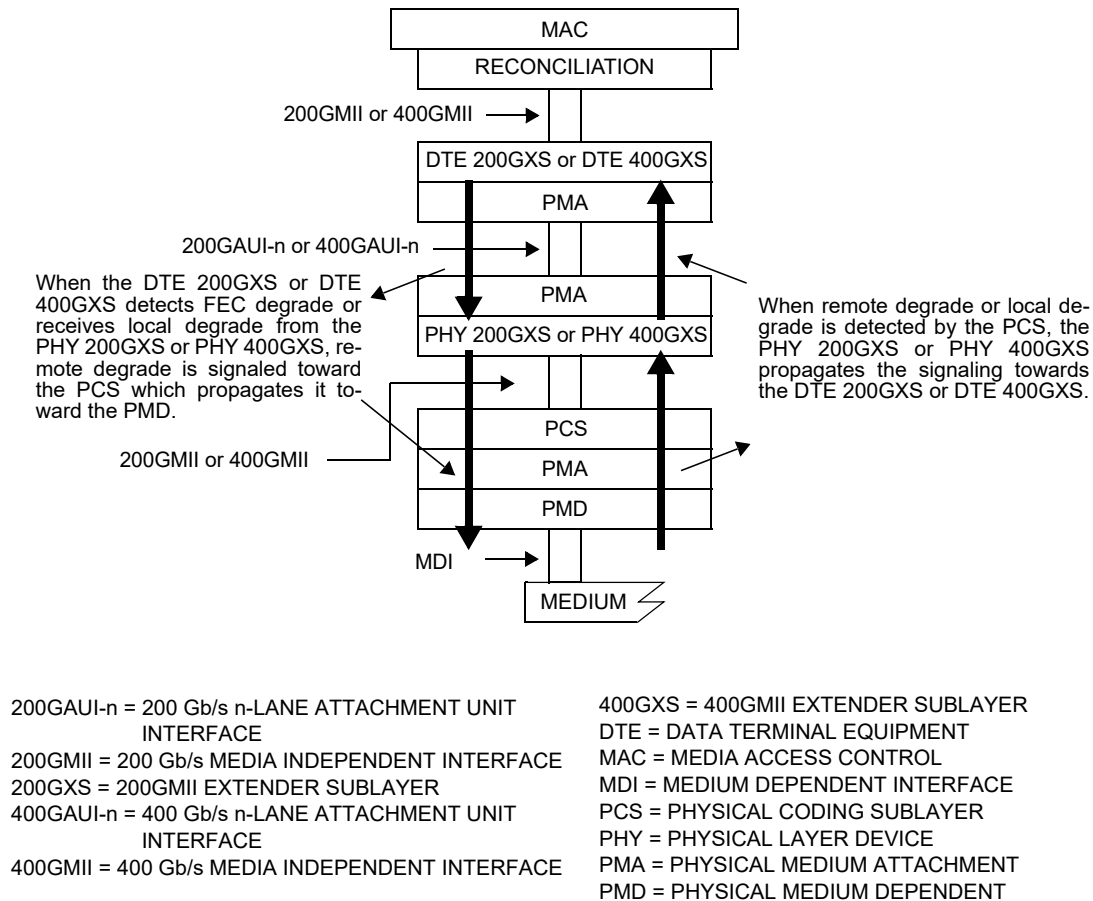


Figure 116-8—Remote Degradate signaling with Extender Sublayer

116.8 Protocol implementation conformance statement (PICS) proforma

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 117 through Clause 124, and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. A PICS is included at the end of each clause as appropriate. Each of the 200 Gigabit and 400 Gigabit Ethernet PICS conforms to the same notation and conventions used in 21.6.